PROPENSITY SCORE STRATIFICATION

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This lecture incorporates material originally developed by Guanglei Hong

Causal Inference

Rubin's Causal Model is a framework for statistical analysis the considers the *potential outcomes* of a *treatment*.

The treatment does not have to be an intervention (e.g. a drug or program). It can be something that occurs naturally (e.g. losing a job, getting married, changing schools).

Causal Inference

There are two potential outcomes.

- One outcome if the person receives the treatment: $Y|D=1$, or Y_1
- One outcome if the person does not receive the treatment: $Y|D=0$, or Y_0

The difference between these two outcomes is the causal effect of the treatment D:

$$Y_1 - Y_0$$

Causal Inference

The outcome we see in reality is the *observed* outcome.

The outcome that did not occur, but could have, is the *counterfactual* outcome.

*One key feature of this framework is that causal estimates are only meaningful if the person could have reasonably been in either the treatment or control groups.

Causal Inference with Regression

Regression based on observational studies will give you a causal estimate of there are no unobserved confounding variables (that predict the treatment and the outcome *conditional on* observed covariates in your model).

But this is a strong assumption, and cannot be empirically verified.

Causal Inference with Regression

Much of modern statistical causal inference is designed to strengthen our argument that there are no unobserved confounding variables.

- Randomized controlled trials (experiments)
- Quasi-experimental designs
  - Difference-in-differences (“diff-in-diff”)
  - Instrumental variables (IV)
  - Interrupted time series
  - Regression discontinuity (RD)
- Propensity score-based methods

Propensity-Score Based Methods

Tries to replicate what we would see in a RCT with observational data.

- Propensity score stratification
- Propensity score matching
- Inverse probability of treatment weighting (IPTW)
- Marginal mean weighting and stratification

The end goal is to have treatment and control groups that are substantively identical based on a large amount of covariates (as opposed to controlling for a large amount of covariates).

Propensity-Score Based Methods

Propensity-score based methods are primarily used for **binary treatments**.

Propensity-score based methods for continuous treatments exist, but are considerably more complicated and not ubiquitous in social sciences.

Example: Kindergarten Retention

Research Question:

What is the effect of retaining a child in kindergarten versus promoting the child?

Causal Estimand:

The average retention effect for the at-risk population.

Non-experimental data:

Early Childhood Longitudinal Study-Kindergarten cohort

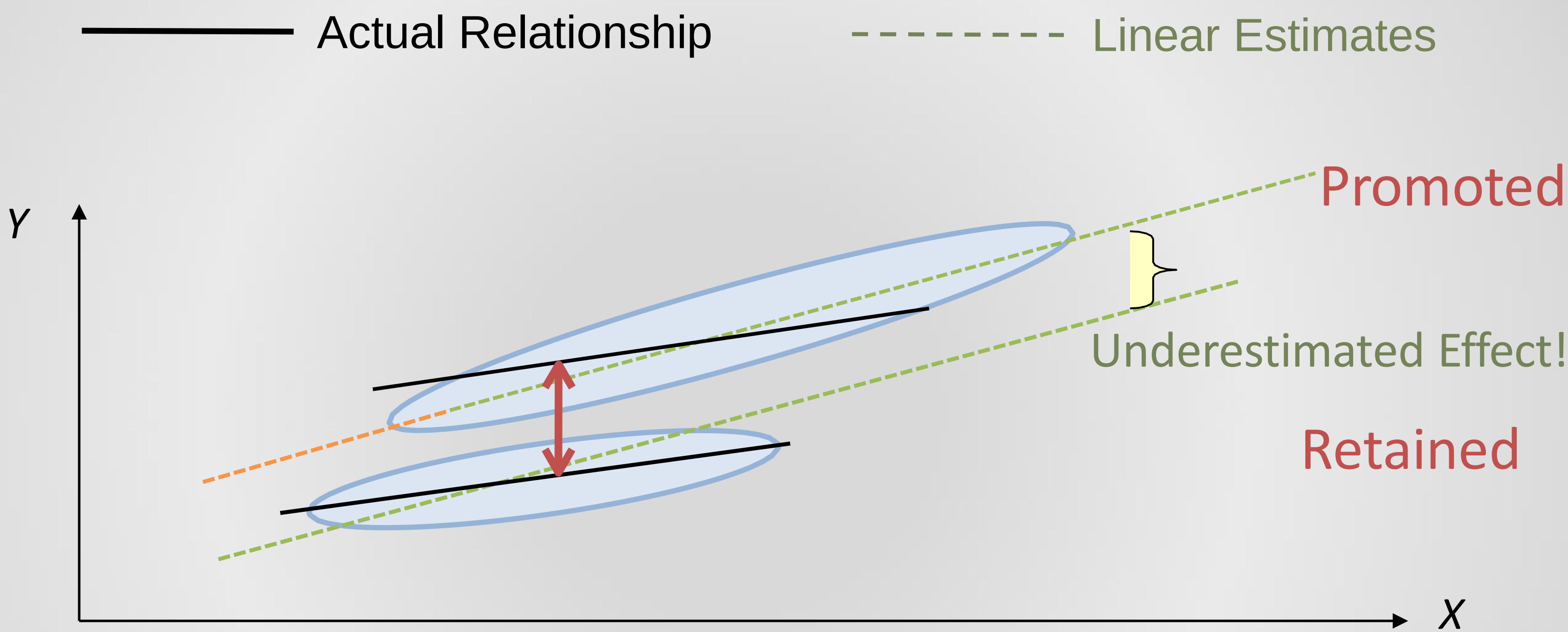
(ECLS-K, 1998-1999) 471 retained; about 12,000

promoted

The Handling of Disturbing Variables (Cochran, 1965)

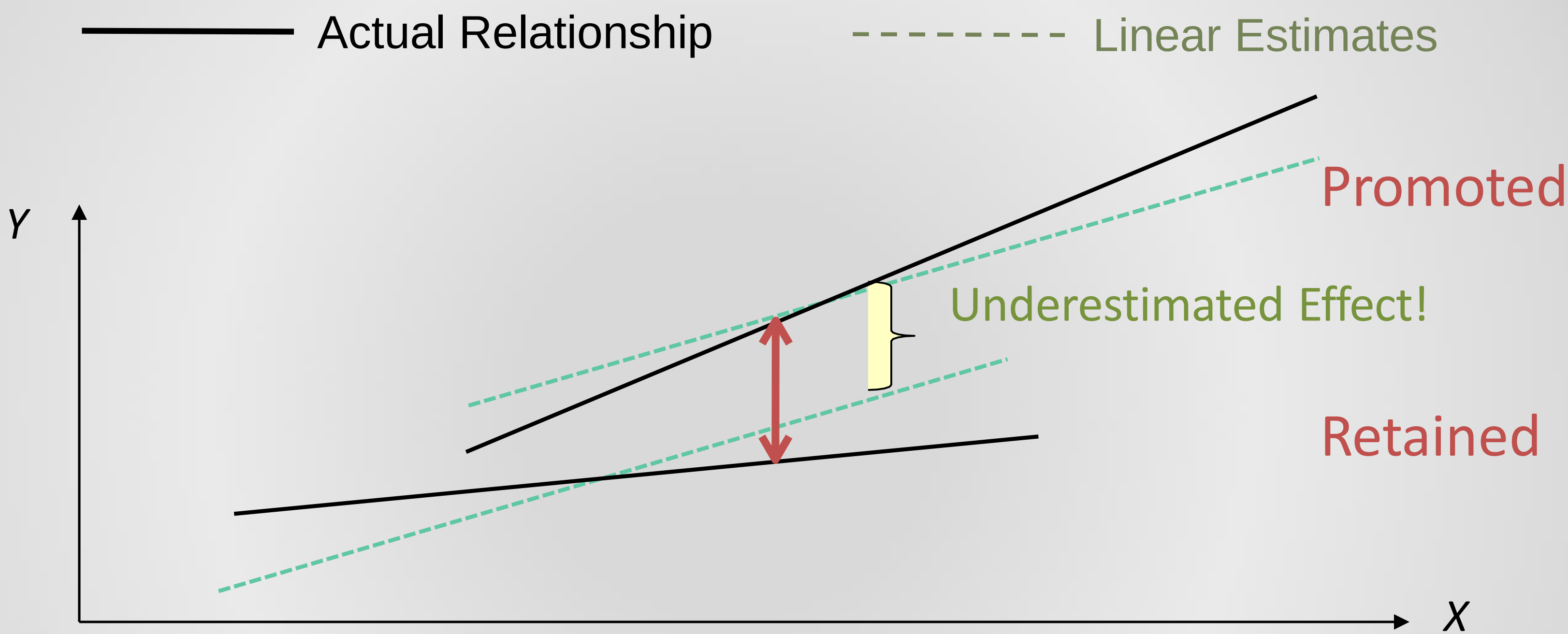
- Adjustment methods
  - Covariance adjustment (OLS, GLM)
    - may rely on linear extrapolation
    - may not detect heterogeneous treatment effect
    - may overlook nonlinearity
    - challenges when X is high-dimensional
  - Matching
    - Works especially well if there is a large control reservoir
  - Subclassification/stratification
    - 90% of bias removed with 5 sub-classes
  - Matching or subclassification with covariance adjustment

Linear Extrapolation



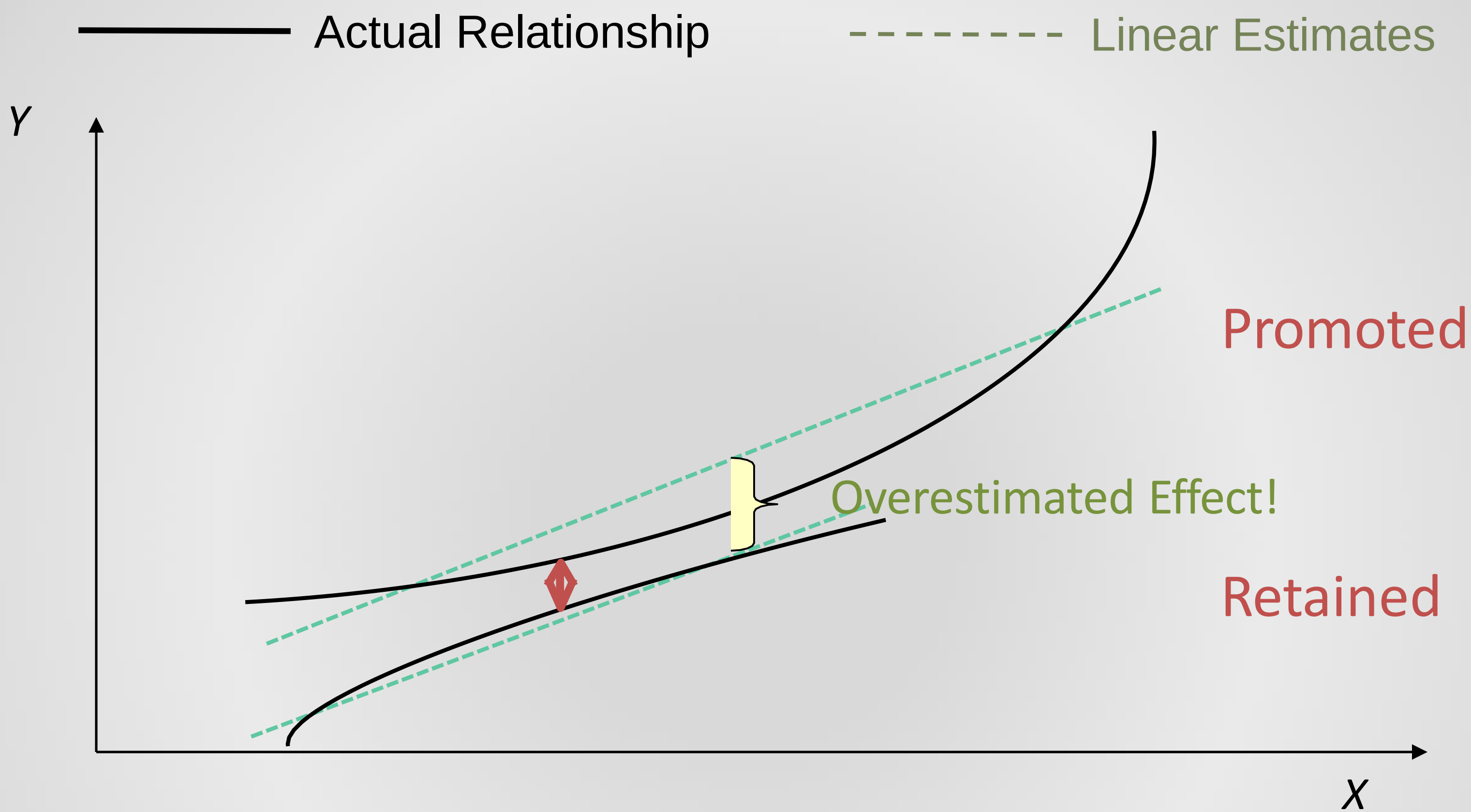
- High-achieving promoted children have a overly strong leverage on the slope estimation
- Extrapolated result for low-achieving retainees without empirical support

Bias if Overlooking Linear Interaction



- If the two treatment groups do not have the same mean of X, omitting the interaction between *X* and *D* will bias the estimate of the average treatment effect

Bias If Overlooking Nonliearity



- If the two treatment groups do not have the same mean and variance of X, omitting the nonlinear relationship between X and Y will bias the estimate of the average treatment effect.

Propensity Score

A unidimensional index that summarizes all the observed pretreatment information that predicts treatment assignment

Property:

The joint distribution of **X (all confounding covariates)** is balanced between the treatment group and the control group when the propensity score is held constant (Rosenbaum & Rubin, 1983, 1984).

Advantages of Matching and Stratification/Subclassification

- Non-parametric in nature – avoid possible misspecification of the relationship between X and Y|D

- Identifying empirical support for causal inference and avoiding extrapolation

Analytic Procedure for Propensity Score Stratification

1. Collect a large set of pre-treatment covariates **X**
2. Build a logistic regression model that uses **X** to predict *D*; save the estimated propensity score *P* (probability of treatment *D*) in logit units (USE HLM)
3. Compare the distribution of the logit of *P* between the treated group and the control group; identify the common support
4. Stratify the sample on the logit of *P*, start with 5 or 6 strata
5. Check balance, first on the logit of *P* itself, then on each *X*
6. Revise the propensity score model or re-stratify if necessary. Repeat until you have balance.
7. Estimate the treatment effect on the outcome controlling for propensity score strata (USE HLM)
8. (Optional) Examine treatment-by-strata interaction effects.
9. (Optional) Conduct sensitivity analysis: Could a plausible violation of the conditional independence assumption alter the conclusion?

Generalized Linear Model: Logistic Regression

Sampling distribution : $D_i \sim Bernoulli(P_i)$

Logit link function:

$$\ln\left(\frac{P_i}{1-P_i}\right) = \eta_i.$$

Log odds, Logit

Here

$$P_i = pr(D_i = 1 | X_{1i}, X_{2i})$$

Structural model:

$$\eta_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} \ldots$$

Step 2: Specify a Propensity Score Model

- Variable selection for the propensity score model (Brookhart et al, 2008)
  – Including strong predictors of the outcome even if they do not predict the treatment increases precision without increasing bias
  – Including strong predictors of the treatment that do not predict the outcome *reduces* precision without reducing bias

Step 2: Specify a Propensity Score Model

- Functional form of the propensity score model
  - Consider including X^2 and X if the treated group and the control group differ in the variance of X
  - Consider including the product of X_1 and X_2 (interaction terms) if their correlation differs between the treated group and the control group
  - Misspecifying the functional form of the propensity score model is less consequential than misspecifying the functional form of the outcome model when using matching or stratification (Drake, 1993)

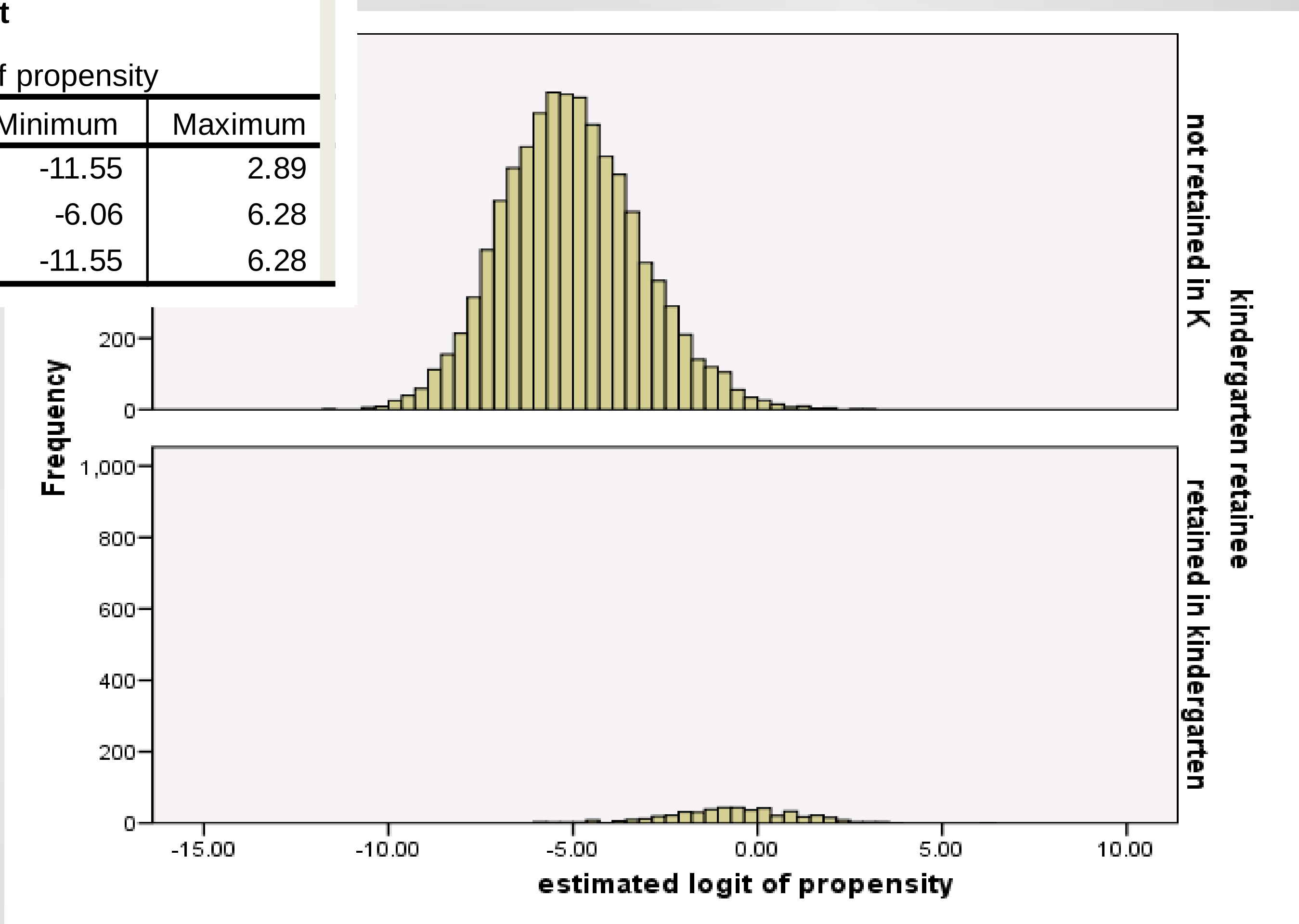
Step 3: Identify Common Support

Example: Kindergarten Retention

Report

estimated logit of propensity

kindergarten retainee	Minimum	Maximum
not retained in K	-11.55	2.89
retained in kindergarten	-6.06	6.28
Total	-11.55	6.28



Step 4: Stratify on the Propensity Score

	Retained			Promoted		
Stratum	*N*	*Mean*	*SD*	*N*	*Mean*	*SD*
$L = 0$	0	---	---	3,087	-7.14	0.85
$L = 1$	9	-5.42	0.40	3,054	-5.38	0.39
$L = 2$	12	-4.25	0.24	1,661	-4.27	0.25
$L = 3$	14	-3.42	0.20	983	-3.50	0.20
$L = 4$	12	-2.98	0.10	323	-2.97	0.10
$L = 5$	24	-2.57	0.17	441	-2.55	0.16
$L = 6$	23	-2.15	0.07	154	-2.14	0.07
$L = 7$	47	-1.75	0.15	213	-1.77	0.15
$L = 8$	48	-1.24	0.14	143	-1.27	0.14
$L = 9$	46	-0.83	0.11	85	-0.86	0.10
$L = 10$	48	-0.46	0.12	49	-0.47	0.11
$L = 11$	47	-0.01	0.11	35	-0.04	0.15
$L = 12$	49	0.53	0.22	16	0.56	0.17
$L = 13$	45	1.16	0.20	8	1.14	0.17
$L = 14$	39	1.99	0.33	3	1.80	0.17
$L = 15$	8	3.70	1.09	0	---	---

Additional example of Balance from Jean (2016)

	Treated			Untreated			
Stratum	**N**	**Mean**	**SD**	**N**	**Mean**	**SD**	**t-stat**
All	**2940**	**2.62**	**.393**	**5406**	**2.19**	**.35**	**51.26**
1	11	**1.67**	0.20	452	**1.71**	0.24	**-0.62**
2	11	**1.87**	0.32	453	**1.89**	0.25	**-0.31**
3	6	**2.05**	0.17	226	**1.98**	0.23	**0.73**
4	13	**2.08**	0.36	219	**2.02**	0.24	**0.88**
5	17	**2.05**	0.31	446	**2.06**	0.25	**-0.07**
6	34	**2.12**	0.23	430	**2.10**	0.27	**0.43**
7	46	**2.18**	0.23	418	**2.19**	0.24	**-0.26**
8	24	**2.17**	0.35	208	**2.22**	0.26	**-0.83**
9	37	**2.23**	0.27	195	**2.22**	0.25	**0.37**
10	78	**2.22**	0.30	385	**2.26**	0.25	**-1.40**
11	16	**2.27**	0.24	100	**2.27**	0.23	**0.10**
12	22	**2.33**	0.30	94	**2.29**	0.24	**0.63**
13	37	**2.32**	0.29	79	**2.32**	0.23	**-0.00**
14	31	**2.31**	0.24	85	**2.29**	0.26	**0.49**
15	110	**2.33**	0.24	354	**2.36**	0.23	**-0.96**
16	151	**2.38**	0.26	312	**2.37**	0.24	**0.38**
17	203	**2.44**	0.25	261	**2.44**	0.22	**-0.04**
18	244	**2.49**	0.25	220	**2.47**	0.24	**0.85**
19	88	**2.58**	0.25	66	**2.51**	0.24	**1.81**
20	93	**2.59**	0.30	62	**2.53**	0.26	**1.30**
21	104	**2.58**	0.28	51	**2.55**	0.21	**0.61**
22	147	**2.59**	0.23	84	**2.64**	0.23	**-1.76**
23	172	**2.63**	0.28	60	**2.55**	0.24	**1.88**
24	377	**2.67**	0.27	87	**2.69**	0.28	**-0.75**
25	420	**2.79**	0.28	44	**2.80**	0.30	**-0.32**
26	448	**3.11**	0.34	15	**3.17**	0.40	**-0.67**

Step 5: Check Balance in Covariates

- Hypothesis testing vs. standardized bias computation
  - Hypothesis testing: no statistically significant mean difference between the treated group and the control group in 95% of the covariates after stratification
  - Standardized difference: mean difference in each covariate between the treated group and the control group should ideally be <.10.

Step 5: Check Balance in Covariates

Hypothesis testing based approaches:

- Two-way ANOVA (general linear model) with each covariate as the outcome of treatments and propensity strata
  - The F ratio of the main effect of treatments close to 1
  - The F ratio of the interaction between treatments and propensity strata close to 1 (see Rosenbaum and Rubin, 1984, Figures 1 and 2)
- Alternatively, t-tests (PS and continuous covariates) or chi-square tests (for categorical covariates) for each stratum-by-treatment interaction.

  *This can be useful for diagnosing problematic strata.
- Make sure at least 95% of you tests are non-significant. This indicates overall balance.

Step 5: Check Balance in Covariates

Standardized mean difference (SMD)

$$\frac{\bar{x}_{Treated} - \bar{x}_{Control}}{\sqrt{(\sigma^2_{Treated} + \sigma^2_{Control})/2}}$$

- Older literature suggests a SMDs should be under .25 for all or nearly all covariates (Stuart, 2010; Stuart and Rubin, 2007)
- Recent literature suggests that standardized mean differences under .10 are recommended (Austin, 2011; Nguyen, Collins, Spence et al., 2017; Zhang, Kim, Lonjon, and Zhu, 2019)

Step 6: Re-stratify If Necessary

You can try…

- Dividing the sample into a larger number of strata

- Subdividing some strata in which there is a lack of balance between the treated group and the control group

- Combining strata

❖ If none of the above seem productive, revise the propensity score model and start over

Step 7: Compute Within-Stratum Mean Difference in the Outcome

$$Y = \beta_0 + \beta_1 D + \sum_{s=2}^{S} \beta_s L_s + \varepsilon$$

β_1 is your treatment effect,
if your assumptions are true

Step 8: Propensity Score Subclassification in Combination with Model-Based Adjustment

$$Y = \beta_0 + \beta_1 D + \sum_{s=2}^{15} \beta_s L_s + \beta_{16}\eta + \varepsilon$$

Controlling for the logit of the propensity score removes residual bias and improves precision

$$\hat{\beta}_1 = -8.00, \; se = 0.60, \; t = -13.34$$

$$\varepsilon \sim N(0, \sigma^2)$$

- Only for a continuous outcome (step 8 different for a binary outcome)
- May include other strong predictors of the outcome (e.g., pretest).
- Useful for moderation analysis. For example, if interested in differential effects by gender, include gender and its interaction with *D* (treatment) in the above model.

Steps 9: Sensitivity Analysis

1. Speculate the existence of an unobserved confounder, U.
2. The bias attributable to the omission of U is $\pi(E\{U|D=1\} - E\{U|D=0\})$, where π is the partial regression coefficient for U in predicting Y within each treatment group.
3. Empirically identify the largest bias = $\pi(E\{U|D=1\} - E\{U|D=0\})$ that is plausible.
4. Remove the hypothetical bias, $\pi(E\{U|D=1\} - E\{U|D=0\})$, from the estimate of the treatment effect.
5. If the confidence interval for the new estimate of the treatment effect leads to a change of the decision about the null hypothesis, then the inference is sensitive to the omission of U.

Analytic Procedure for Propensity Score Matching

1. Collect a large set of pre-treatment covariates **X**
2. Build a logistic regression model that uses **X** to predict *D*; save the estimated propensity score *P* in logit units
3. Compare the distribution of the logit of *P* between the treated group and the control group; identify the common support.
4. Match each treated unit with a control unit on the logit of *P*, e.g., nearest neighbor matching without replacement.
5. Check balance within matched pairs first on the logit of *P* itself, then on each *X*
6. Revise the propensity score model or modify the matching if necessary to achieve balance
7. Compute the average within-pair difference in outcome between treatment groups, with possible adjustment for prognostic covariates
8. Conduct sensitivity analysis

Strengths and Limitations of Propensity Score Stratification

- Strengths
  - Removes upwards of 90% of bias in all observed covariates
  - Robust to misspecifications of the functional form of the propensity score model
  - Easy to identify empirical support for causal inference
  - Easy to examine heterogeneity of treatment effect
  - Easy to estimate the treatment effect on the treated
- Limitations
  - Inconvenient with multiple treatments or continuous treatments
  - Unfeasible for handling time-varying covariates

Assumptions for Causal Inference

- Strong ignorability
  - No unobserved covariates that affect both the treatment and outcome, conditional on covariates
  - Cannot be verified, but sensitivity analysis estimates how robust estimates are to violation

- Stable Unit Treatment Value Assumption (SUTVA)
  - There is only one version of the treatment
  - No interference between units (spillover effects) – treatment of one unit does not affect the outcomes of another